

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Third Semester of B.Tech. (CE/IT) Examination
May 2011

IT201 Database Management

Date: 21/05/2011

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately

SECTION – I

Q1

- | | | |
|-----|--|---|
| (a) | List and explain significant difference between file processing system and DBMS. | 4 |
| (b) | Define the following terms.
1. Atomicity 2. Candidate Key
3. Super key 4. Assertions | 4 |
| (c) | What should be the properties of a transaction? Explain. | 3 |

Q2

- | | | |
|-----|---|---|
| (a) | Compare entity set with Weak Entity Set. Also explain generalization. | 6 |
| (b) | Explain normalization. Also discuss BCNF and 4NF. | 4 |
| (c) | How multi valued attribute can be represented in table. | 2 |

OR

Q2

- | | | |
|-----|---|---|
| (a) | Construct an E-R diagram for university registrar's office. The office maintains data about each class, including the instructor, the enrollment, and the time and place of the class meetings. For each student-class pair, a graded is recorded. Document all assumptions that you make about the mapping constraints | 6 |
| (b) | List difference between DDL and DML | 3 |
| (c) | Why Hash structure not the best choice for a search key on which range queries are likely? | 3 |

Q3

- | | | |
|-----|--|---|
| (a) | Consider the following relational database
Branch = (branch-name, branch-city, assets)
Customer = (customer-name, customer-street, customer-city)
Account = (branch-name, account#, balance)
Depositor = (customer-name, account#)
Loan = (branch-name, loan#, amount)
Borrower = (customer-name, loan#) | 9 |
|-----|--|---|

Specify the following queries in relational algebra and in SQL

- | | | |
|-------|--|---|
| (i) | Names of customers with both accounts and loans at Perryridge branch. | |
| (ii) | Average balance of all customers in Harrison having at least 2 accounts. | |
| (iii) | Names of branches having at least one account, with average account balances. | |
| (b) | With following set of functional dependencies
$A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C, AC \rightarrow D$
Compute an irreducible set of FD that is equivalent to this given set | 3 |

OR

Q3

- (a) Consider the following relational database
Department(DeptNo, DeptName)
Employee(Eno, Ename, Job, Mgr, Hiredate, Sal, Comm, DeptNo);
Specify the following queries in relational algebra and in SQL.
(i) List the names, jobs of all clerks in deptno 8.
(ii) List the details of all employee who have the managers
(iii) Display all employees who were hired during 1991.
- (b) Define the terms Functionally Dependent and Fully Functionally Dependent with appropriate example.

SECTION – II

Q4

- (a) Explain conflict Serializability. Why is it preferred? 4
(b) Explain the stages of query processing with diagram. 4
(c) Explain UPDATE and DELETE SQL statement. 3

Q5

- (a) Explain divide and conquer strategy with example. 5
(b) Explain the following terms 5
(i) Deadlock (iii) Integrity (v) Relation
(ii) Grant (iv) Canonical form
- (c) A restriction of projection can be transform into projection of a restriction. Justify your answer. 2

OR

Q5

- (a) What are the two approaches of data security? Explain both the data approaches with example. 5
(b) Write the different types of locks and their use in concurrency control with suitable examples. 7

Q6 Answer the following questions (Any Four)

- 1) Explain in brief ACID properties of a database transaction. 12
2) Define data abstraction. Explain the three level of abstraction using the three-tier architecture proposed by ANSI/SPARC
3) Explain what is meant by the terms Candidate Key and Composite Key.
4) Explain system recovery with example.
5) Explain the concept of request modification with example.